

Advanced Robotics

ISSN: 0169-1864 (Print) 1568-5535 (Online) Journal homepage: www.tandfonline.com/journals/tadr20

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To cite this article: M. Zorjan, V. Hugel, P. Blazevic & B. Borovac (2015) Influence of rotation of humanoid hip joint axes on joint power during locomotion, Advanced Robotics, 29:11, 707-719, DOI: 10.1080/01691864.2015.1013057

To link to this article: https://doi.org/10.1080/01691864.2015.1013057

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Advanced Robotics, 2015
Vol. 29, No. 11, 707–719, http://dx.doi.org/10.1080/01691864.2015.1013057

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FULL PAPER

Influence of rotation of humanoid hip joint axes on joint power during locomotion

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(*Received 14 July 2014; revised 4 November 2014; accepted 12 January 2015*)

Most of the fully actuated legs of existing humanoids have a straight kinematics, i.e. roll, pitch and yaw joint rotary axes are aligned, respectively, along the longitudinal, lateral and vertical axes of the robot's body. This work explores different orientations of the hip joint axes where the yaw joint and the pitch joint axes are rotated about the longitudinal axis, keeping the serial order – yaw, roll and pitch – from the pelvis. Two walking movements are simulated dynamically: forward walk and turn-in-place. The measurements of joint mechanical power (JMP) consumption are used to compare the different kinematics configurations. The results obtained from simulation show that the 45° inward and outward rotations of hip yaw and pitch joint axes about the longitudinal axis bring a significant reduction of pitch JMP, and lead to a better JMP distribution among the hip joint actuators compared with the classical straight configuration. In addition, the 45° outward configuration presents a significant reduction of JMP consumption of the hip pitch joint of the supporting leg during acceleration and regular steps, compared with the 45° inward configuration.

Keywords: hip kinematics; legged locomotion; humanoid modelling; mechanical power

1. Introduction

Conventional humanoid robots have less degrees of freedom (DOFs) than humans, the human body being a highly redundant system. Hence, an artificial imitation of man has to be significantly simplified in both the number of DOFs and the design of some complex structures. However, the differences in the design can have significant influence on the humanoid capabilities. Among the design aspects, the DOF configuration of the hip is an interesting case of study.

The present-day human-sized robots are capable of performing sustainable dynamic walking. Most of them have the same kinematics of the legs: starting from the pelvis, the order of the joints is hip yaw, hip roll, hip pitch, knee pitch, ankle pitch and ankle roll. All Roll, pitch and yaw joint axes are parallel to the x longitudinal, y lateral and z vertical axes of the robot's body, respectively (see Figure 1 for hip joint axes). Such joint layouts will be referred to as 'usual' or 'classical'. Examples of such humanoids are P2, P3 and ASIMO [1,2] from HONDA research group, HRP-2P [3,4] from the Japan National Institute of Advanced Industrial Science and Technology and Kawada Industries, Inc. (HRP-2P has a specific cantilever structure at the hip), H6 and H7 [5,6] from the University of Tokyo, WABIAN [7,8] from Waseda University (this robot has a 2-DOF waist combination of a roll and a yaw joint that are attached in the middle between the hip joints), KHR-2 [9] from Korea Advanced Institute of Science and Technology. In the matter of hip layout, Okada et al. [10] placed the hip joints at the waist with all vertical, horizontal and lateral axes intersecting at the same point, which leads to increased workspace and energy saving thanks to a backlash clutch at the knees.

All the above mentioned humanoids have a classical straight orientation of the hip joint axes. However, in the case of Johnnie [11] from the Technical University of Munich and the small size biped robot NAO from the French Aldebaran Robotics company, the three joint axes are not mutually orthogonal because the hip yaw joint axes are inclined inward in the frontal plane. In the case of Johnnie, the yaw joint inclination is 15° (Figure 2(a)). According to the authors [12], this leads to a better power distribution among hip joints. In the case of NAO, the hip yaw joints are coupled and inclined by 45° inward (Figure 2(b)), and only one motor is required to drive both yaw joints. This allows reducing building costs and saving space in the lower part of the trunk.[13] The ARMAR-4 humanoid [14] also features a specific design of the hips (Figure 2(c)), where the new hip-joint axes configuration results from two rotations starting from the usual straight kinematics layout. The yaw and pitch axes are first rotated inward about the roll axis by 45°, then the pitch and roll axes are rotated forward about the 45° inclined yaw axis by 45°. In this way, all three

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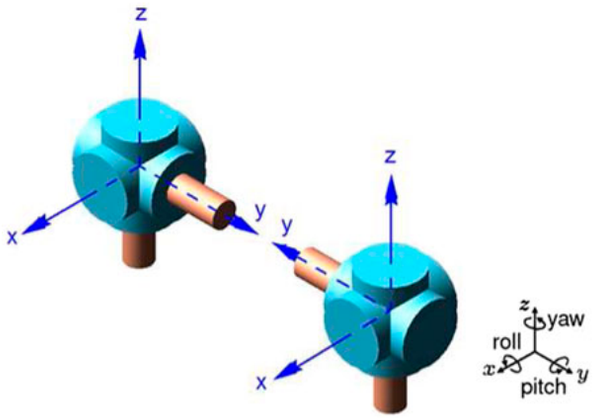


Figure 1. Usual hip joints orientation.

joint actuators may contribute to generate higher torques for some movements such as pitch movements. However, there is no reported study about the effectiveness of this configuration yet.

Humanoids with unconventional design of the hips are not common, and the question that arises is the following: is there any real benefit to change the design at the hips compared with the usual design? We investigated the influence of a change of the hip kinematics on joint mechanical power (JMP) consumption for several kinds of motion. Starting from the usual hip design model with a set of three mutually orthogonal rotary joints (Figure 1), the hip kinematics was changed by rotating the hip joint coordinate frame (HJCF) constituted by the joint rotation axes. Previous studies involved a rotation about the vertical z-axis and a rotation about the longitudinal x-axis that were applied to the hips of a simulated humanoid programmed to execute knee flexion/extension, swaying hips, forward/backward pelvis motion,[15] and knee flexion with additional load in the hands.[16] The studies showed that the rotation of hip joint axes about the x-axis can bring benefits in the matter of JMP distribution. However, the kinds of motion above are not enough to conclude on the JMP distribution of a humanoid robot's hips because they do not include locomotion moves, which are among the dominant activities. That is why this paper focuses on two kinds of dynamic walking gaits, forward and turn-in-place, featuring leg swing and stance phases. The JMP is measured along two walking paths – forward and turn –, and is compared between the different hip configurations. Rotations of the HJCF of 15°, 30°, 45° and 60° in the clockwise and counterclockwise directions are used for the comparison. In addition, the walking gait is decomposed in terms of JMP, torque and angular velocity to explain the origin of better JMP distribution at the hips.

The paper is structured as follows. Section 2 deals with the robot's general model that is used for simulation, and the model of contact between the ground and robot's feet. Section 3 describes the motions used in this study: forward walk and turn-in-place. Results about the JMP required at the hip joints with different hip kinematics are presented and compared in Section 4. Section 5 is devoted to the discussion of the results. Conclusions are given in Section 6.

2. Humanoid robot model

2.1. Description

The robot was modelled and simulated in 3D with Adams dynamics software from MSC.Software. The robot's characteristics match the ROMEO robot from Aldebaran company. It has 38 DOFs, is 1.38 m tall and weighs 35.54 kg. Table 1 gives the main characteristics of the ROMEO humanoid that were adopted for the simulation. Each leg has 3 DOFs at the hip, 1 DOF at the knee, 2 DOFs at the ankle and 1 DOF at the toe. The toe joints remain fixed during simulation, therefore each foot is considered as a single link. The kinematic structure of the ROMEO robot is presented in Figure 3. The order of the leg joints from the hip to the foot is the classical one.

2.2. Model of ground-foot contact

Each foot sole is equipped with eight semi-spheres that make contact with the ground. Each semi-sphere is associated with one sensor for measuring the corresponding normal ground reaction force (GRF). The sum of all readings from the sensors constitutes the total GRF. The positions of the semi-spheres below the foot sole are shown in Figure 4.

The normal force F_n between the sphere and the ground is essentially modelled as a non-linear spring damper, using the following *impact* model [17]:

$$F_n = kp^e + c(p)\mathrm{d}p/\mathrm{d}t \tag{1}$$

where:

- p stands for the penetration of one geometry into another, here the sphere into the flat ground,
- k is the stiffness,
- $\mathrm{d}p/\mathrm{d}t$ is the penetration velocity at the contact point,
- e is a positive real value denoting the force exponent,
- $c(p)$ is the damping coefficient as a function of penetration.

$c(p)$ is expressed as the step function:

$$\begin{aligned} p \le 0 &\Rightarrow c(p) = 0 \\ 0 < p < p_{\max} &\Rightarrow c(p) = c_{\max}\left(\frac{p}{p_{\max}}\right)^2\left(3 - 2\frac{p}{p_{\max}}\right) \\ p_{\max} \le p &\Rightarrow c(p) = c_{\max} \end{aligned}$$

where $p_{\max}$ is a positive real value specifying the boundary penetration where to apply the maximum damping $c_{\max}$.

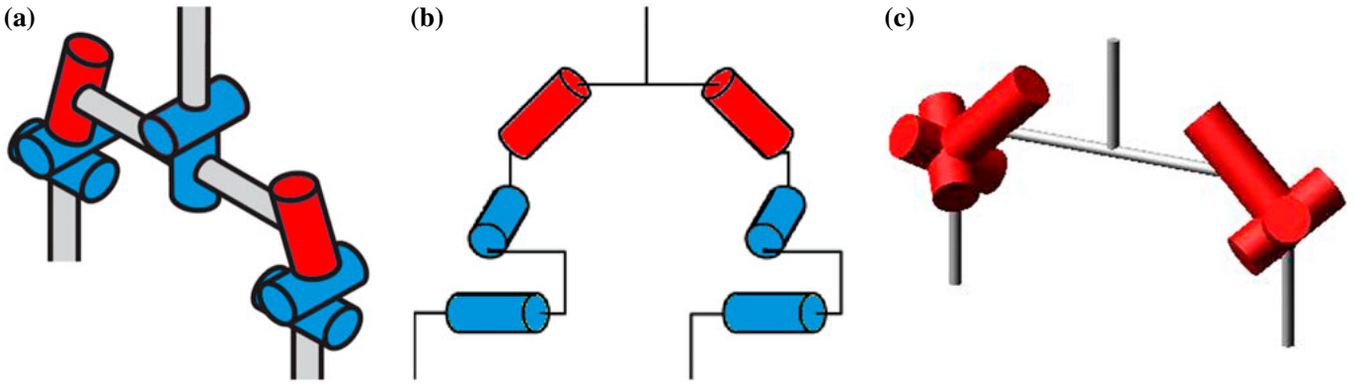


Figure 2. Hip kinematics: (a) Johnnie, (b) NAO and (c) ARMAR-4.

Table 1. Data for the robot's body parts. M_T = total mass, H_T = total height.

Part	Mass [Kg]	Length [m]	DOF
Head	0.04 M_T	0.19 H_T	1
Arms (2)	0.26 M_T	0.4 H_T	14
Trunk	0.3 M_T	0.35 H_T	9
Legs (2)	0.4 M_T	0.46 H_T	14
Total	$M_T = 35.54$	$H_T = 1.38$	38

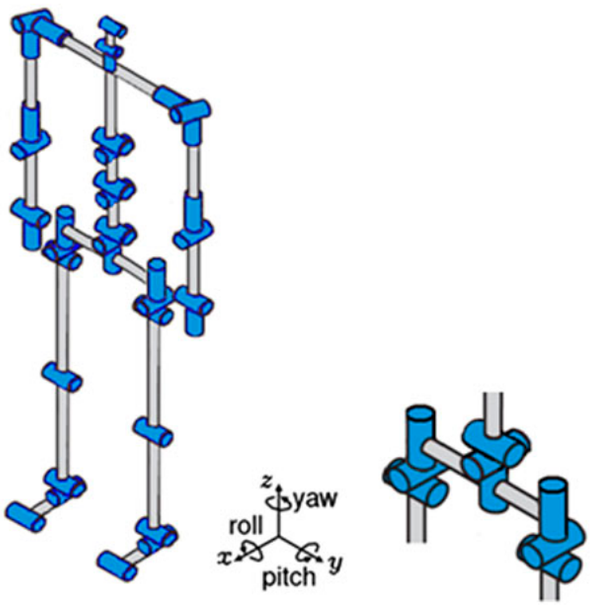


Figure 3. Kinematic structure of the ROMEO humanoid robot.

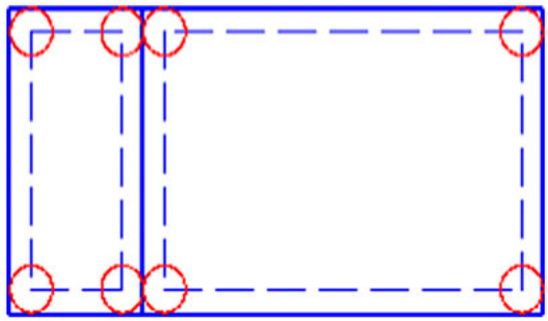


Figure 4. Position of the sensors below the foot sole (length: 0.25 m, width: 0.13 m).

3. Simulation setup and experiments

The simulation with the dynamic software is useful to check the robustness and the balance keeping of the walk, and the consequences of little variations of the foot-sole horizontality due to acceleration/deceleration moves. It is also useful to analyse the influence of the parameters of the ground-interaction model on the quality of the walk, and to simulate the entire control chain, as it will be implemented on a real robot. The model of the robot in the dynamics software is generated automatically thanks to Matlab script files that read an input file that contains all the characteristics of the robot. This allows saving development time for the model design when modifications are to be taken into account, especially modifications of the kinematics of the hips.

The trajectories of the desired angular positions and velocities related to the motion planned are calculated by inverse kinematics in Matlab and sent to the dynamics software every 10 ms. The joint trajectory tracking is achieved through PID controllers inside the dynamic software. The inputs of each controller are the position and velocity of the related joint. The ouput is the torque applied to the motor associated with the joint.

The LIP-3D model (linear inverted pendulum) [18] coupled with the zero moment point (ZMP) technique [19–21] is used to design the walking patterns used for our study.[22] The humanoid robot is modelled as a 3D linear inverted pendulum. ZMP points for each support foot are selected to generate the trajectory of the centre of mass (COM) that takes the shape of successive hyperbolic curves with continuity of velocity. Angle joints are then calculated through inverse kinematics from the trajectory of the COM. The COM is considered to be fixed with respect to the torso, which is a reasonable approximation in the

case of the LIP-3D model. We designed a specific inverse kinematics solution that is adapted to the change of joint axis orientation.[23] This technique is based on geometrical calculations to get knee coordinates, and on the decomposition of inverse leg kinematics, i.e. inverse kinematics at the hip for the one part, and inverse kinematics at the ankle for the other part. This calculation assumes that the leg has six rotary degrees of freedom, three at the hip, one at the knee and two at the ankle. The orientation of rotary joints at the hip and at the ankle can be arbitrary.

In our experiments, the height of the LIP-3D model was set to 0.69 m; this means that the COM is concentrated at this height in the standing posture.

The main criterion that can help to evaluate the influence of the different hip joint axis orientation is the average JMP, namely P_{avg}. This power does not take into account electrical power. Torques were measured at the joints. The instantaneous JMP $P(t)$ is related to the torque $\tau(t)$ and the angular velocity $\omega(t)$ at each joint as:

$$P^i(t) = \tau^i(t) \cdot \omega^i(t) \tag{2}$$

where i is the ordinal number of the joint.

The average JMP P_{avg} is calculated as the average of absolute values of JMP over the period under consideration:

$$P_{\text{avg}} = < |P(t)| >_T \tag{3}$$

3.1. *Forward walk*

The initial position (0) is the upright posture with both legs stretched. Before starting to walk the robot flexes the knees until reaching an angle of 52°; this position is referred to as position (1) with reference time $t = 0$. This position corresponds to a flexing ratio of 0.9 of the hip height to the total length of the stretched leg. Then, the robot walks 0.6 m forward in 2.4 s by executing four steps. A step is defined as the distance between two consecutive impacts of legs. In our experiment each step is 0.2 m. The walking forward motion is decomposed into the following 6 phases (Figure 5 with the left leg as *starting* leg), each phase lasts 0.4 s, and the four successive impacts match positions (3)–(6):

- (1) → (2), $t \in [0.0, 0.4]$, outward hip sway followed by inward hip sway to initiate a lateral velocity of the COM to prepare the first step.
- (2) → (3), $t \in [0.4, 0.8]$, first step of 0.2 m executed by the *starting* leg in the swing phase, while the *other leg* is in support. The COM advances 0.1 m.
- (3) → (4), $t \in [0.8, 1.2]$, second step, the *starting* leg is in support while the *other* leg executes a swing phase with a stride of 0.4 m, twice the step length. The COM advances 0.2 m.
- (4) → (5), $t \in [1.2, 1.6]$, third step, the *starting* leg executes a swing phase with a stride of 0.4 m – twice the step length –, while the *other* leg is in support. The COM advances 0.2 m.
- (5) → (6), $t \in [1.6, 2.0]$, fourth step, the *starting* leg is in support and the *other* leg executes a swing phase with a stride of 0.2 m. The COM advances 0.1 m.
- (6) → (7), $t \in [2.0, 2.4]$, outward hip sway followed by inward hip sway to stop the motion of the COM.

The forward walking simulation is run with the left leg as the *starting* leg, and repeated with the right leg as the *starting* leg.

3.2. *Turn-in-place*

In this experiment, the robot turns 90°, left or right, by executing six rotary steps of 30° each. The step in this motion is defined as the angle between two consecutive foot impacts of the same leg. During each step, the robot's trunk turns 15°. The robot starts to flex the knees, then executes an outward followed by an inward hip sway, like in the forward walking. The step time is the same as in the forward walk, i.e. 0.4 s.

The turn-in-place walking can be decomposed as follows:

- (1) → (2), $t \in [0.0, 0.4]$, outward hip sway followed by inward hip sway to initiate a lateral velocity of the COM to prepare the first step.
- (2) → (3), $t \in [0.4, 0.8]$, first step, the *starting* leg executes a swing phase with a turning angle of 30°, while the *other* leg remains in support.
- (3) → (4), $t \in [0.8, 1.2]$, second step, the *starting* leg is in support and the *other* leg executes a swing phase with a turning angle of 30°. At the end of this phase both feet are parallel.
- (4) → (5), $t \in [1.2, 1.6]$, same as (2) → (3).
- (5) → (6), $t \in [1.6, 2.0]$, same as (3) → (4).
- (6) → (7), $t \in [2.0, 2.4]$, same as (2) → (3).
- (7) → (8), $t \in [2.4, 2.8]$, same as (3) → (4).
- (8) → (9), $t \in [2.8, 3.2]$, outward hip sway followed by inward hip sway to stop the motion of the COM.

The turn-in-place walking simulation is run with the left leg as the *starting* leg, and repeated with the right leg as the *starting* leg. Figure 6 shows the decomposition of the left turn-in-place used for this 2nd walking pattern experiment,

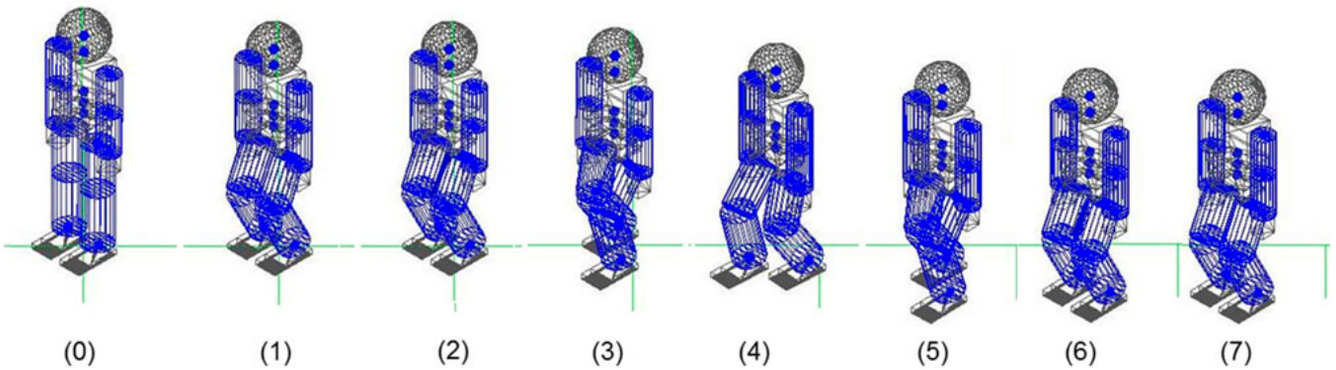


Figure 5. Forward walk: (0) initial position, (1) knee-flexed position, (2) position after outward/inward hip sway, (3) position after 1st step with the left foot, (4) position after 2nd step with the right foot, (5) position after 3rd step with the left foot, (6) position after 4th step with the right foot, (7) final position after outward/inward hip sway. Position (7) is the same as position (1).

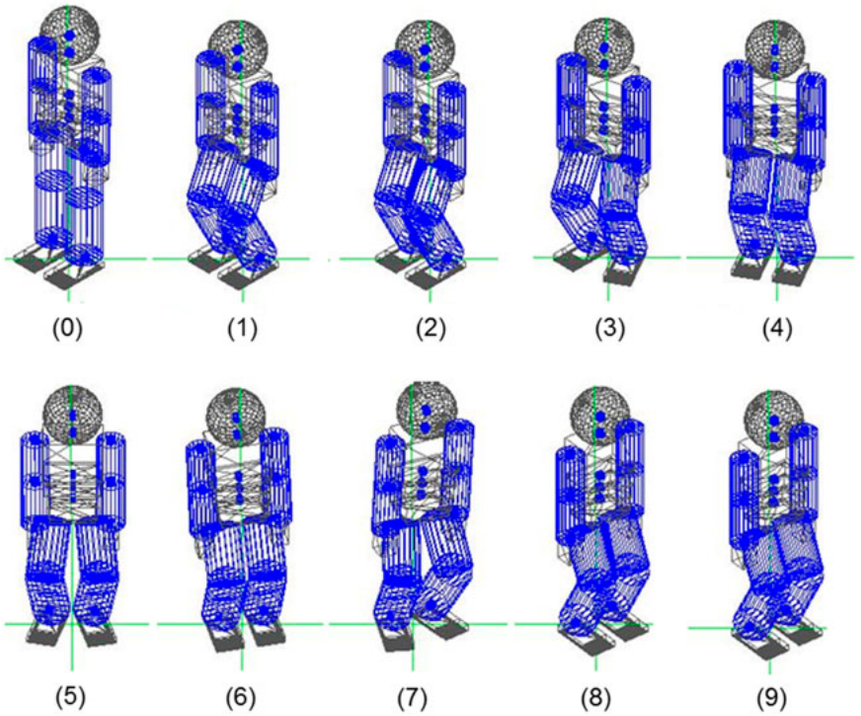


Figure 6. Left turn-in-place of 90°: (0) initial position, (1) knee-flexed position, (2) position after outward/inward hip sway, (3) position after outward rotary step of 30° with the left foot, (4) position after inward rotary step of 30° with the right foot – the robot puts its right foot parallel with the left foot, (5)(7) same as (3), (6)(8) same as (4), (9) final position after outward/inward hip sway. Position (9) is the position (1) rotated by 90° about the vertical passing through the middle of the feet in the counterclockwise direction.

the *starting* leg being the left leg. In the case of right turns, the *starting* leg is the right leg.

3.3. Hip kinematics used in simulations

The hip joint was modelled as a set of three mutually orthogonal single rotary joints whose axes of rotation constitute a coordinate frame. This frame was inclined outwards and inwards as a whole by rotation about the longitudinal x-axis. Angles of rotation used are 15°, 30°, 45° and 60°. Figure 7 shows the outward and inward configurations after 45° rotation of the HJCF.

4. Results

4.1. Forward walk

Results presented in this section are obtained by considering the simulation with the left leg as starting leg and the simulation with the right leg as starting leg, and by calculating the average of signals in both cases.

Figure 8 displays the average consumption for both ankles, both knees, both hips and both legs (ankles, knees and hips) in the different configurations of rotated yaw and pitch joint axes. The JMP consumed at the hips is larger than the JMP consumed at the knees or the JMP consumed at the

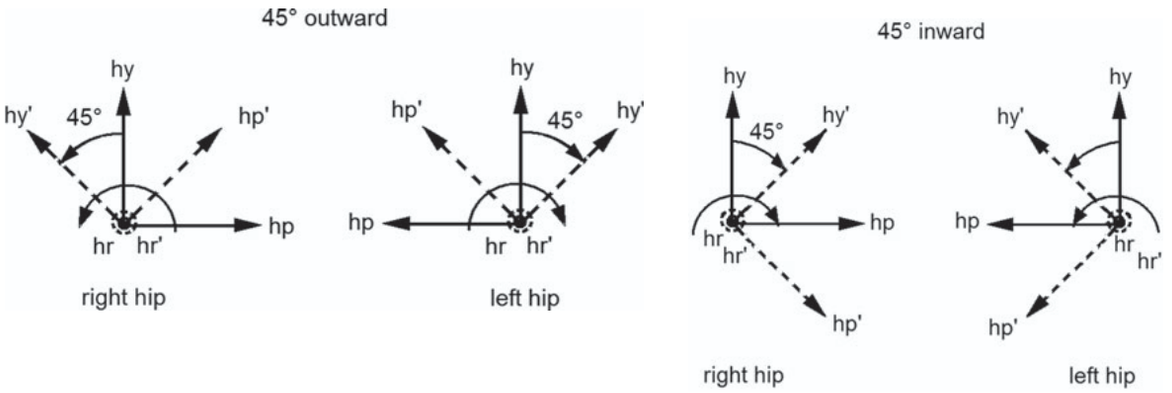


Figure 7. HJCF rotated 45° outward and inward about the x-axis. hy, hr and hp stand for hip yaw, hip roll and hip pitch, respectively.

ankles, and accounts for nearly half of the cumulated JMP of all leg joints. Whereas the JMP at the knees and the ankles remains nearly the same, the JMP consumption of the hip joints slightly varies among the different configurations.

Figure 9 illustrates the decomposition of hip JMP consumption among the three joints. The following results can be observed:

(1) The roll JMP does not vary so much among the configurations when compared with JMP variations of yaw and pitch joints.
(2) There is a significant JMP reduction at the pitch joint in the 45° and 60° configurations, compared with the JMP in the usual straight configuration. Actually, the pitch JMP in the usual configuration is decreased by 57% in the 45° outward configuration. However, this reduction is at the expense of yaw JMP increase.
(3) The 45° configurations present a better JMP distribution among the three joints.
(4) The pitch JMP in the outward configuration is significantly less than the pitch JMP in the inward configuration of the same angle.

In order to determine the origin of the JMP difference between both 45° configurations, the evolution of JMP over time along the forward walking path is analysed. Figures 10 and 11 depict the JMP, the angular velocity and the actuator torque for the hip joints along the phases of the walking path. Figure 10 refers to the starting leg, and Figure 11, to the other leg. The walking motion is decomposed into six time intervals. During the first interval [0, 0.4] the robot's legs are in double support, and sway the hips to prepare the first step. Then, the starting leg executes a swing phase followed by a stance phase twice, while the other leg makes a stance phase followed by a swing phase twice. The last

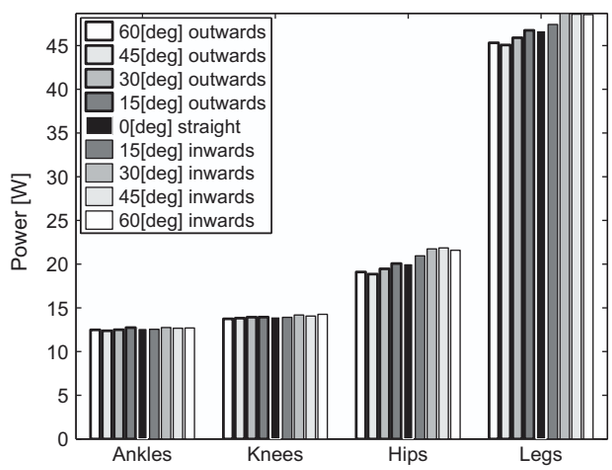


Figure 8. Total average JMP in forward walk for both ankles, both knee, both hip joints and both legs (ankles + knees + hips). Kinematics configurations include outward and inward rotations about the x-axis of the HJCF: from left to right, 60°, 45°, 30°, 15° outwards, 0° (usual straight configuration), 15°, 30°, 45° and 60° inwards.

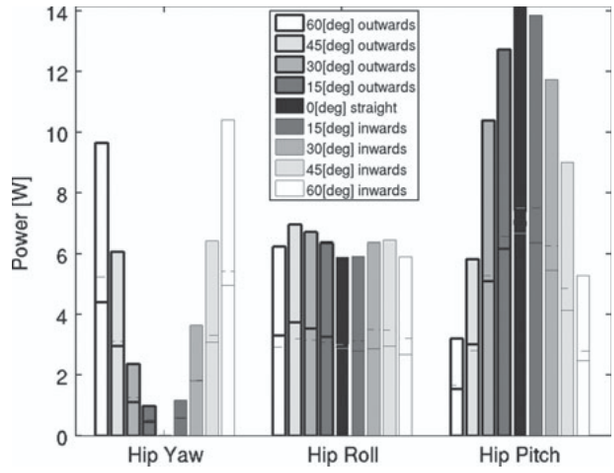


Figure 9. Average JMP in forward walk for yaw joints, roll joints and pitch joints of both hips. The intermediary values refer to the JMP values of hip joints belonging to the starting leg (solid lines), and to the JMP values of hip joints belonging to the other leg (dotted lines). Kinematics configurations include outward and inward rotations about the x-axis of the HJCF: from left to right, 60°, 45°, 30°, 15° outwards, 0° (usual straight configuration), 15°, 30°, 45° and 60° inwards.

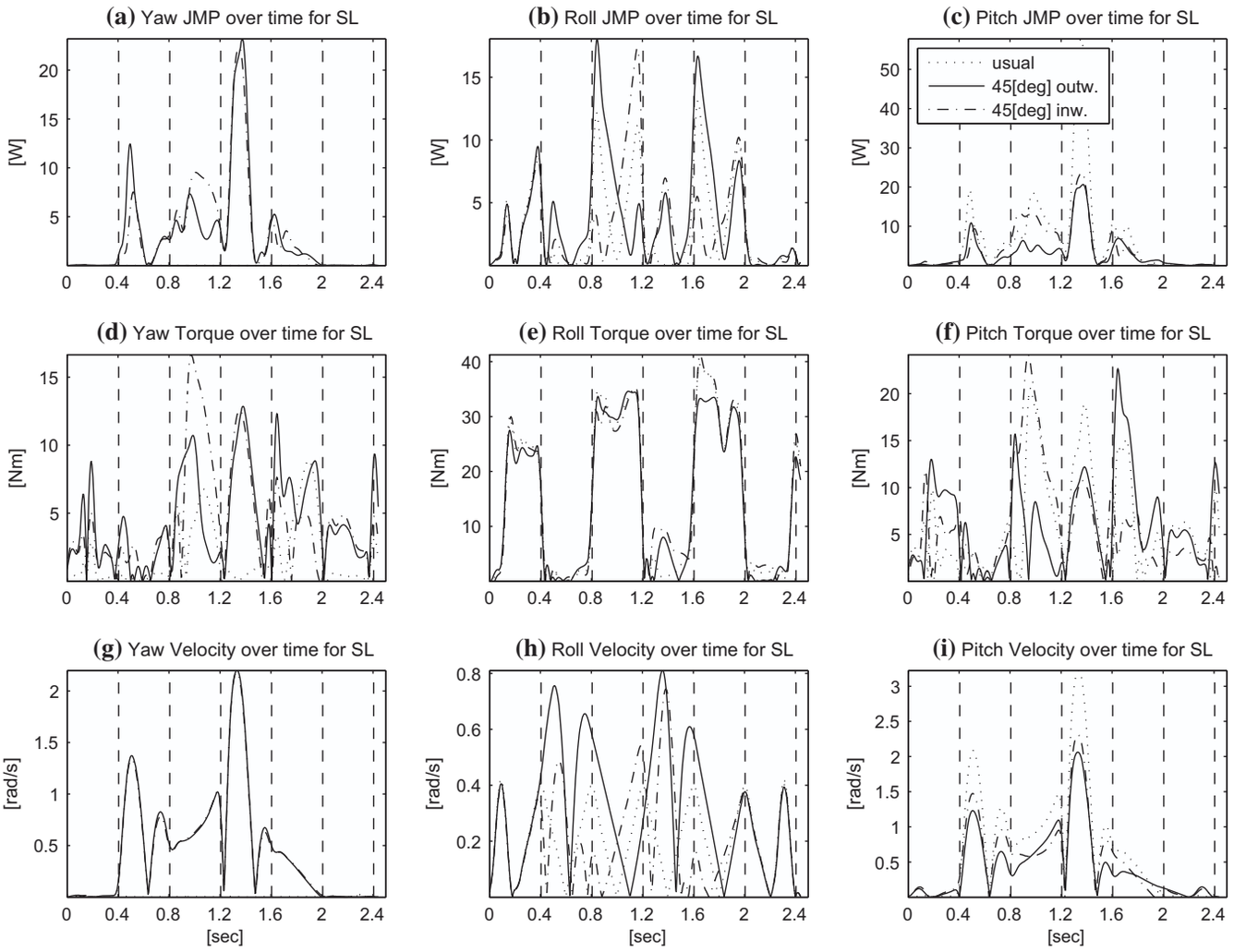


Figure 10. Comparison of the JMP, torque and velocity (absolute values) of the hip joints related to the starting leg SL all along the forward walking path between the usual, the 45° inward and the 45° outward configurations. The first column refers to the yaw joint, the second column to the roll joint and the third column to the pitch joint. [0, 0.4]: hip sway, [0.4, 0.8]: swing, [0.8, 1.2]: stance, [1.2, 1.6]: swing, [1.6, 2.0]: stance, [1.6, 2.0]: hip sway.

time interval [2.0, 2.4] is spent to sway hips to get back to the initial posture with both legs on the ground.

First, the following observations for the *swinging legs* can be made:

- According to Figure 10(c) in [0.4, 0.8] and [1.2, 1.6], and Figure 11(c) in [0.8, 1.2] and [1.6, 2.0], the largest JMP reduction at the pitch joint in the inclined configurations ranges between 50% and 64%, compared with the usual straight configuration. The maximal reduction of pitch JMP occurs when the leg is swinging in the regular steps[1] ([1.2, 1.6] for the starting leg on Figure 10, and [0.8, 1.2] for the other leg on Figure 11) where the pitch JMP reaches a peak due to highest angular velocity. This result can be explained by the fact that the pitch torque and the pitch angular velocities are significantly reduced in the 45° configurations. The amount of JMP is similar for both 45° inward and outward configurations, with smallest values for the roll joints (Figures 10(b) and 11(b)) because of reduced actuated torque.
- The yaw JMP (Figures 10(a) and 11(a)) also reaches a peak in the regular steps for the 45° configurations, whereas it is zero for the usual configuration since the yaw joint velocity is zero.

Second, the following observations related to the support leg can be made:

- Along the acceleration step (Figure 11(c) in [0.4, 0.8]), and along regular steps (Figure 10(c) in [0.8, 1.2], and Figure 11(c) in [1.2, 1.6]), the pitch JMP is significantly less in the 45° outward

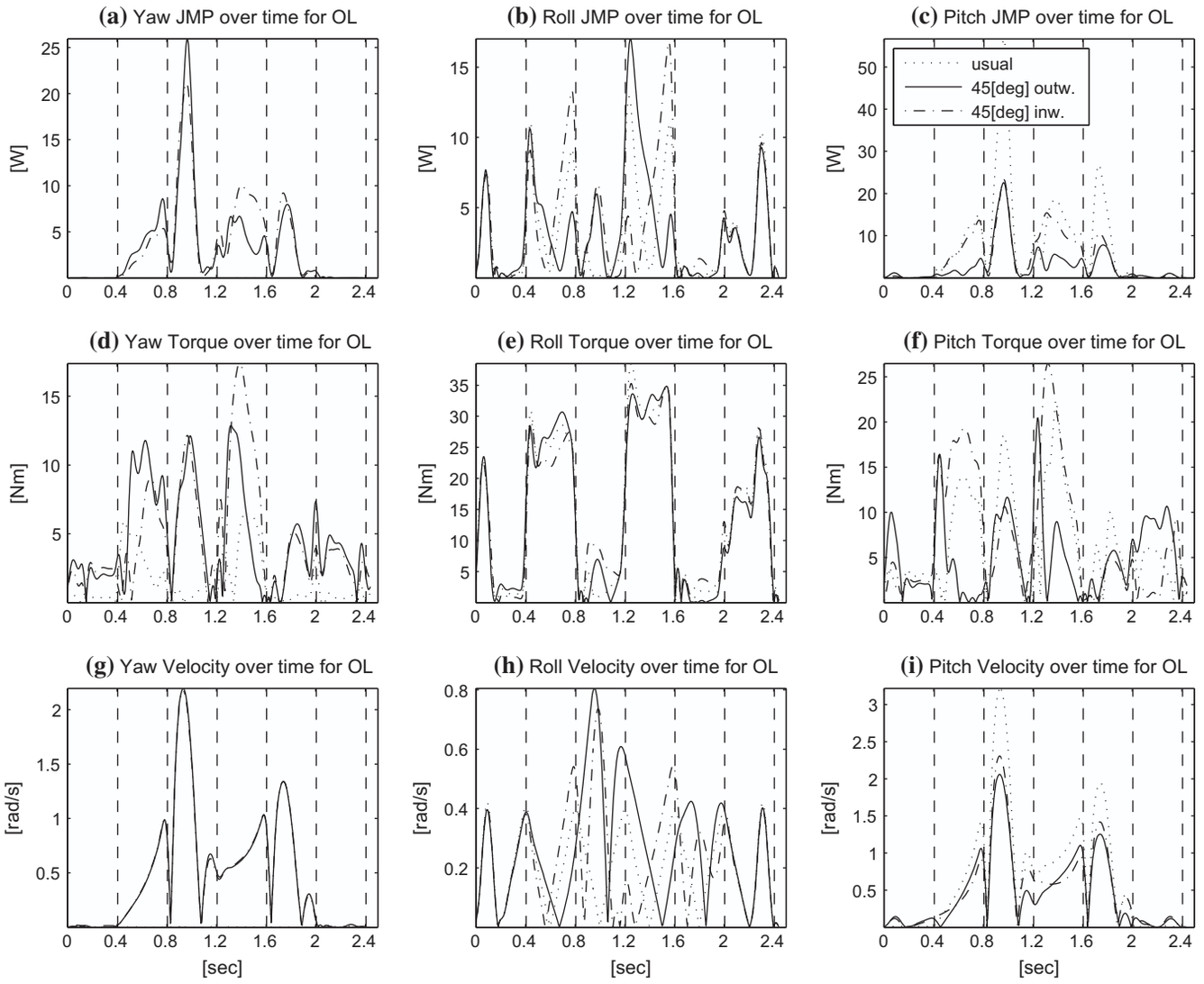


Figure 11. Comparison of the JMP, torque and velocity (absolute values) of the hip joints related to the other leg OL all along the forward walking path between the usual, the 45° inward and the 45° outward configurations. The first column refers to the yaw joint, the second column to the roll joint and the third column to the pitch joint. [0, 0.4]: hip sway, [0.4, 0.8]: stance, [0.8, 1.2]: swing, [1.2, 1.6]: stance, [1.6, 2.0]: swing, [1.6, 2.0]: hip sway.

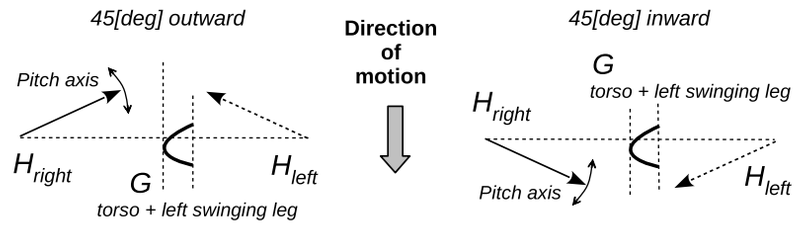


Figure 12. Horizontal projection of pitch axis and trajectory of COM of humanoid's upper part and left swinging leg. Right leg in support. Left: 45° outward configuration. Right: 45° inward configuration.

configuration than in both the usual configuration and the 45° inward configuration. Obviously, this difference of JMP comes from the difference of torques. Actually, the actuated torque in the outward configuration is significantly less than in the inward configuration over the same periods of time (Figure 10(f) in [0.8, 1.2], and Figure 11(f) in [0.4, 0.8] and [1.2, 1.6]).

- Along the deceleration step executed by the starting leg over the period [1.6, 2.0] (Figure 10) to end

the walking motion, this is the opposite, i.e. this time the pitch torque (Figure 10(f) in [1.6, 2.0]) is significantly less for the 45° inward configuration, compared with the outward configuration, therefore leading to a greater amount of JMP in the outward configuration.

The torque reduction at the pitch joint of the supporting leg in the outward configuration over the period [0.4, 1.6] can be explained by considering the decomposition of the actuated torque denoted by τ_P:

$$|\tau_P| = |\dot{L}_H - M_g| \tag{4}$$

where $\dot{L}_H$ is the time derivative of the angular momentum about the pitch axis of the mass composed of the robot's upper body part and the swinging leg, and M_g is the moment due to gravity about the pitch axis of the same mass. In the case of the 45° outward configuration, the COM of the robot's upper part and the swinging leg is located ahead of the pitch axis (Figure 12), whereas it is located behind the pitch axis in the 45° inward configuration. This is due to the knee flexion that requires the yaw joint to turn, which makes the pitch joint axes rotate backward in the 45° outward configuration, and forward in the 45° inward configuration.

After leg impacts at times 0.4, 0.8 and 1.2 s, the hip pitch joint actuator has to provide enough torque for the forward motion acceleration of the robot's upper part and the swinging leg, and for the compensation of the gravity about the pitch axis. In the case of the 45° outward configuration, $\dot{L}_H$ and M_g are the same sign, which explains why the active torque τ_P of the support leg is decreased during the acceleration step and the regular steps. In the 45° inward configuration, $\dot{L}_H$ and M_g are the opposite sign; therefore, the active torque must be increased to cope with both effects. At 1.6 s, the body is decelerated to stop the motion, and the pitch joint actuator of the 45° outward configuration must deliver more torque to deal with both the motion acceleration and the gravity moment.

4.2. Turn-in-place

Since 45° configurations present the most interesting results in forward walk, we only present turn-in-place results related to these configurations and to the usual configuration.

According to the results presented in Figure 13, legs consume nearly twice less JMP than in forward walk. The JMP rate due to the hips is also more important in the turn-in-place walk. This is because the turns request more job from the hip roll joints in comparison with the other hip joints. This is confirmed by the decomposition of JMP consumption among hip joints illustrated in Figure 14. This figure also shows that the rotated configurations do not bring any reduction of JMP in the hip roll joint, and therefore JMP consumption cannot be distributed among the three joints.

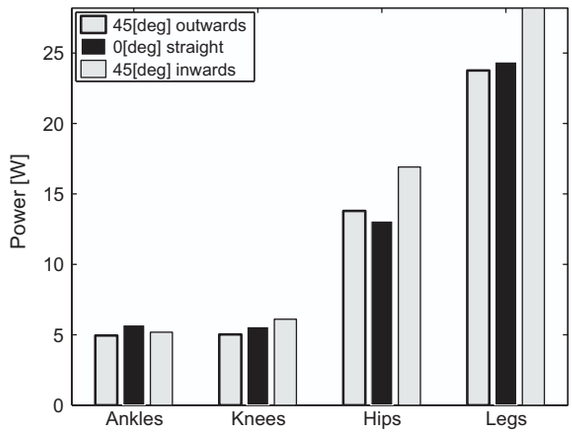


Figure 13. Total Average JMP in turn-in-place walk for ankle, knee and hip joints. Kinematics configurations include outward and inward rotations about the x-axis of the HJCF: from left to right, 45° outwards, 0° (usual straight configuration) and 45° inwards.

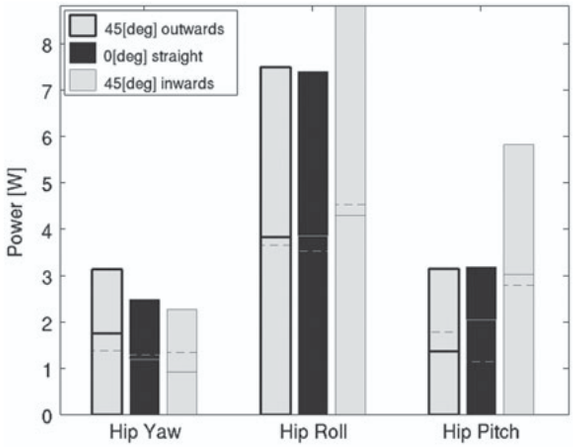


Figure 14. Average JMP in turn-in-place walk for yaw joints, roll joints and pitch joints of both hips. The intermediary values refer to the JMP values of hip joints belonging to the starting leg (solid lines), and to the JMP values of hip joints belonging to the other leg (dotted lines). Kinematics configurations include outward and inward rotations about the x-axis of the HJCF: from left to right, 45° outwards, 0° (usual straight configuration) and 45° inwards.

The rotation of hip joint axes does not bring any advantage in the turn-in-place walks, but does not bring any drawback either especially in the 45° outward configuration. In addition, the simulations show that the total leg JMP in turn-in-place remains nearly half less than the JMP in forward walk. Therefore, the turn-in-place walking gaits cannot be considered as decisive in the choice of better suited rotated configuration.

5. Discussion

In this study, the hip joint axes were rotated as a whole to keep the orthogonality. However, in some existing robots like Johnnie and NAO, only the yaw joint at the hip is

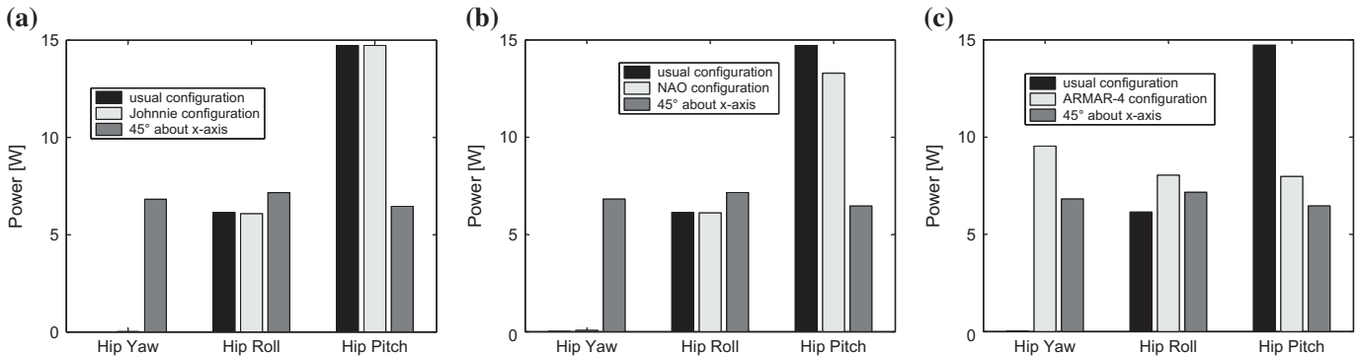


Figure 15. Comparison of the average JMP in forward walking between the usual configuration, the 45° outward configuration: (a) Johnnie robot's hip configuration – 15° inclination of hip yaw axis, (b) NAO robot's hip configuration – 45° inclination of hip yaw axis and (c) ARMAR4 robot's hip configuration – 45° inward rotation of yaw and pitch axes about the roll axis, followed by a 45° forward rotation of pitch and roll axes about the 45° inclined yaw axis.

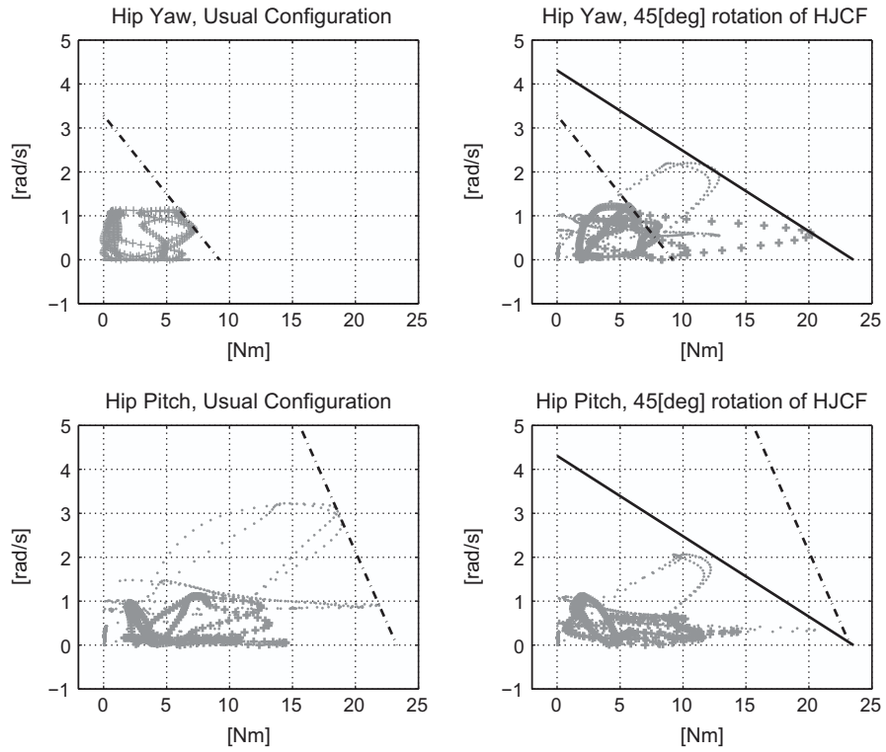


Figure 16. Comparison of angular velocity/torque diagrams of the hip yaw and pitch joints for all simulated moves between usual configuration and 45° HJCF rotation. Plus signs are related to turning motion.

inclined in the frontal plane, while the other two axes of the pitch and roll joints remain unchanged in the horizontal plane. As a consequence, the yaw axis is no more orthogonal to the pitch axis. Figure 15(a) and (b) shows the resulting average JMP related to these configurations compared with the 45° outward configuration and the usual configuration for the task of forward walk, for the Johnnie and the NAO robot models, respectively.

These results show that the inclination of the yaw axis alone does not bring any benefit in the matter of JMP distribution among the joints. Therefore, it is better to keep the orthogonality of the three joints. Orthogonality was kept in the case of the hips of the ARMAR-4 robot. Figure 15(c) shows the resulting average JMP related to the ARMAR-4 compared with the 45° outward configuration and the usual configuration. The hip pitch JMP is actually reduced, and the JMP is better distributed among the hip joints. But the ARMAR-4 configuration appears not to be better than the 45° outward configuration, the yaw JMP being most higher. This means that the inclined configuration must be carefully designed by taking into account the characteristics of the walk that will be applied.

Thanks to the 45° rotation of the HJCF, it is possible to use the same kind of motors for the yaw joint and the

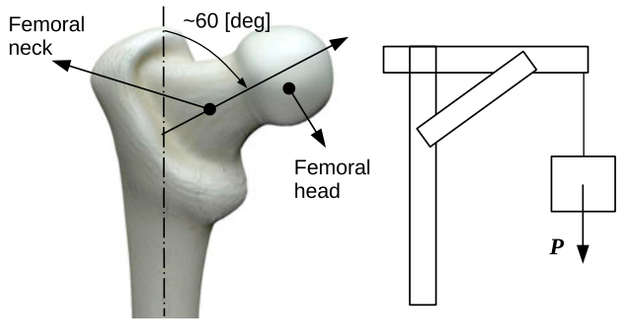


Figure 17. Top part of the human femur compared with a strut used to strengthen a T-shaped structure.

pitch joint. This is illustrated by the diagrams (Figure 16) of angular velocity/torque at the hip yaw and pitch joints for all the simulated moves in the cases of the usual hip joint configuration and the 45° outward configuration.

The walking algorithm used for simulations is based on the LIP-3D model where the trajectory of the COM is calculated from predefined ZMP points. The acceleration of the COM is not continuous at the moments of impacts, and there is no double support phase. It is possible to smooth accelerations by introducing a short double support phase, and to implement a better walking algorithm with some preview controller.[24] However, we did not implement such improvements to keep the worst case where to check the stress on the actuators.

The comparison between the 45° outward and inward configurations shows an advantage to the outward configuration where torque and JMP are reduced along the acceleration and regular steps. This advantage is due to the location of the COM that is ahead of the pitch joint axis in the outward configuration. However, this advantage depends on the amount of knee flexion that is necessary to make the robot walk while keeping a constant height of the COM. In the case of a natural human walk, this advantage disappears since the leg is stretched after its impact on the ground, i.e. there is no knee flexion to lower the COM. If knee flexion is small, the walking steps will be small and the effectiveness of the configuration will be similar to the usual configuration. A larger knee flexion can provide larger steps, which can bring benefits for the acceleration and regular steps, but the COM may be too much lowered, which may increase the risk of fall along the deceleration steps required to stop the motion. Therefore, it is possible to determine the stride length for the regular steps that may be the result of a trade-off between the nominal torque allowed for the actuators and the desired walking speed.

From a biomechanical point of view, the human hip joint is inclined externally in the sagittal plane, and the axis of the femur head/neck makes an angle of approx. 60° with respect to the vertical (Figure 17).[25]

Although the 60° inclination is close to the 45° inclination of the pitch joint proposed in this paper, the inclined shape of the femoral neck is more the result of an evolution process that aimed to distribute the lines of forces at the surface of the femur neck and head in order to resist the vertical and horizontal shear forces exerted on the hip joint.[25] The inclination of the femoral neck plays the role of a strut in a T-shaped structure to better resist the vertical shear forces when the structure has to bear a significant load. The advantage of the 45° pitch joint axis proposed in this paper is that it can resist vertical shear forces better than a horizontal axis. However, the yaw joint axis is 45° inclined too and requires to be strengthened compared with a vertical axis in the classical straight configuration. In addition, the human hip joint is a pure spherical joint whereas the hip joint of the humanoid is composed of successive yaw, roll and pitch joints, which leads to a different distribution of the forces. As a conclusion, the inclination of the hip joints of the humanoid robot can bring benefits in matter of torque distribution among the joints, but the human joint is much more complex, and the inclined shape of the femur neck comes from a lot of factors that would be difficult to take into account to build a fully bio-inspired robotic hip.

6. Conclusion

The results presented in this paper show that the 45° configurations – the hip yaw and the hip pitch joint axes are rotated about the longitudinal axis – allow reducing the JMP at the hip pitch joint significantly in forward walk while producing JMP at the yaw joint. In addition, the 45° rotation of the HJCF lead to a better JMP distribution among the hip joints, which enable to choose similar actuators for the hip yaw and pitch joints. In matter of actuated torque and JMP, the 45° outward rotation is better than the 45° inward rotation for the supporting leg along the acceleration and regular steps, but not along the deceleration steps. This result highlights the importance of the location of the COM relative to the hip joint axes during walking steps. Further investigations will focus on the influence of other kinematics configurations at the ankle joint with respect to locomotion tasks.

Funding

This research was supported by the French Single Inter-Ministry Fund ROMEO. The ROMEO Project is a R& D project led by the French company Aldebaran Robotics. It involved twelve partners, including four SMEs, seven academic partners and one user representative. Half of the 10M€global budget was funded by the French Government, the Ile de France region and the City of Paris.

Notes

1. A regular step is a step where the COM moves with a constant average velocity

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References

[1] Hirai K, Hirose M, Haikawa Y, Takenaka T. The development of Honda humanoid robot. In: IEEE International Conference on Robotics and Automation; 1988; Philadelphia, PA. p. 1321–1326.

[2] Sakagami Y, Watanabe R, Aoyama C, Matsunaga S, Higaki N, Fujimura K. The intelligent ASIMO: system overview and integration. In: IEEE/RSJ International Conference on Intelligent Robots and Systems; 2002; Lausanne, Switzerland. p. 2478–2483.

[3] Kaneko K, Kanehiro F, Kajita S, Hirukava H, Kawasaki T, Hirata M, Akachi K, Isozumi T. Humanoid robot HRP-2. In: IEEE International Conference on Robotics and Automation; 2004; New Orleans, FL. p. 1803–1090.

[4] Fujiwara K, Kanehiro F, Kajita S, Yokoi K, Saito H, Harada K, Kaneko K, Hirukawa H. The first human-size humanoid that can fall over safely and stand-up again. In: IEEE/RSJ International Conference on Intelligent Robots and Systems; 2003; Las Vegas, NV. p. 1920–1926.

[5] Nishiwaki K, Sugihara T, Kagami S, Kanehiro F, Inaba M, Inoue H. Design and development of research platform for perception–action integration in humanoid robot: H6. In: IEEE/RSJ International Conference on Intelligent Robots and Systems; 2000; Takamatsu, Japan. p. 1559–1564.

[6] Nishiwaki K, Kagami S, Kuniyoshi Y, Inaba M, Inoue H. Online generation of humanoid walking motion based on a fast generation method of motion pattern that follows desired ZMP. In: IEEE/RSJ International Conference on Intelligent Robots and Systems; 2002; Lausanne, Switzerland. p. 2684–2689.

[7] Ogura Y, Aikawa H, Shimomura K, Kondo H, Morishima A. Development of a new humanoid robot WABIAN-2. In: IEEE International Conference on Robotics and Automation; 2006; Orlando, FL. p. 76–81.

[8] Lim H, Kaneshima Y, Takanishi A. Online walking pattern generation for biped humanoid with trunk. In: IEEE International Conference on Robotics and Automation; 2002; Washington, DC. p. 3111–3116.

[9] Kim J, Park I, Lee J, Kim M, Cho B, Oh J. System design and dynamic walking of humanoid robot KHR-2. In: IEEE International Conference on Robotics and Automation; 2005; Barcelona, Spain. p. 1443–1448.

[10] Okada M, Shinohara T, Gotoh T, Ban S, Nakamura Y. Double spherical joint and backlash clutch for lower limbs of humanoids. In: IEEE International Conference on Robotics and Automation; 2003; Taipei, Taiwan. p. 491–496.

[11] Löffler K, Gienger M, Pfeiffer F. Sensor and control design of a dynamically stable biped robot. In: IEEE International Conference on Robotics and Automation; 2003; Taipei, Taiwan. p. 484–490.

[12] Gienger M. Entwurf und realisierung einer zweibeinigen laufmaschine [Design and Building of a running biped.]. Frankfurt/Main: VDI-Verlag. 2005. p. 30–31. ISBN 3-18-337801-9.

[13] Gouaillier D, Hugel V, Blazevic P, Kilner C, Monceaux J, Lafourcade P, Marnier B, Serre J, Maisonnier B. Mechatronic design of NAO humanoid. In: IEEE International Conference on Robotics and Automation; 2009; Kobe, Japan. p. 2124–2129.

[14] Asfour T, Schill J, Peters H, Klas C, Bücker J, Sander C, Schulz S, Kargov A, Werner T, Bartenbach V. ARMAR-4: A 63 torque controlled humanoid robot. In: IEEE/RAS International Conference on Humanoid Robots (Humanoids); 2013; Atlanta, GA. p. 390–396.

[15] Zorjan M, Hugel V, Blazevic P. Comparison between different humanoid leg hip kinematics through dynamic simulation. J. Ind. Eng. Manage. 2011;2:131–137.
[16] Zorjan M, Hugel V, Blazevic P. Influence of hip joint axes change of orientation on power distribution in humanoid motion. In: IEEE International Conference on Automation, Robotics and Application (ICARA); 2011; Wellington, New Zealand. p. 271–276.
[17] MSC.Software Adams online documentation, Available from: www.mscsoftware.com. Santa Ana (CA): MSC SoftwareCorporation. 92707.
[18] Kajita S, Matsumoto O, Saigo M. Real-time 3D walking pattern generation for a biped robot with telescopic legs. In: IEEE International Conference on Robotics and Automation; 2001; Seoul, Korea. p. 2299–2306.
[19] Vukobratovic M, Stepanenko J. On the stability of anthropomorphic systems. Math. Biosci. 1972;15:1–37.
[20] Vukobratovic M, Borovac B, Surla D, Stokic D. Biped locomotion, dynamics, stability and application mobile robotics. A practical introduction. 2nd ed. Berlin: Springer-Verlag; 1990.
[21] Vukobratovic M, Borovac B. Zero-moment point? Thirty five years of its life. Int. J. Humanoid Robot. 2004;1: 157–173.
[22] Hugel V, Jouandeau N. Walking patterns for real time path planning simulation of humanoids. In: IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN); 2012; Paris, France. p. 424–430.
[23] Zorjan M, Hugel V. Generalized humanoid leg inverse kinematics to avoid singularities. In: IEEE International Conference Robotics and Automation (ICRA); 2013; Karlsruhe, Germany. p. 4791–4796.
[24] Dimitrov D, Wieber P, Ferreau HJ, Diehl M. On the implementation of model predictive control for online walking pattern generation. In: IEEE International Conference Robotics and Automation (ICRA); 2008; Pasadena, CA. p. 2685–2690.
[25] Kapandji AI. Anatomie fonctionnelle: tome 2 – membre inférieur [Functional Anatomy, Volume 2, Lower Limb.]. 6th ed. Paris: Maloine; 2009. ISBN: 978-2224032142. (in French).